SHA-1 RTL optimization on VTR

Technical report · reproducible open 45 nm FPGA proxy

Three cycle-equivalent RTL transformations reduce estimated composite PPA by 2.27%

Prepared by Göther Labs

Evidence snapshot: 10 August 2026

1. Executive result

CERTIFIED BASELINE VERSUS ACCEPTED RTL

> **The accepted RTL reduces the equal-weight area-delay-active-power score by 2.27% (95% CI: 2.12% to 2.41%) without changing the module interface, cycle-visible behavior, latency or register count.**

Primary metric	Baseline median	Accepted median	Paired improvement	95% CI	State
Total area	16,614,693 MWTA	16,614,693 MWTA	**0.15%**	+0.21% +0.08%	better
Critical path	15.0054 ns	14.2090 ns	**5.20%**	+5.75% +4.64%	better
Active power	9.9125 mW	9.7755 mW	**1.38%**	+1.87% +0.89%	better
Composite PPA	1.000000	0.977335	**2.27%**	+2.41% +2.12%	better

Certified PPA improvement vs. baseline

64 fixed, paired VPR seeds · lower metric values are better

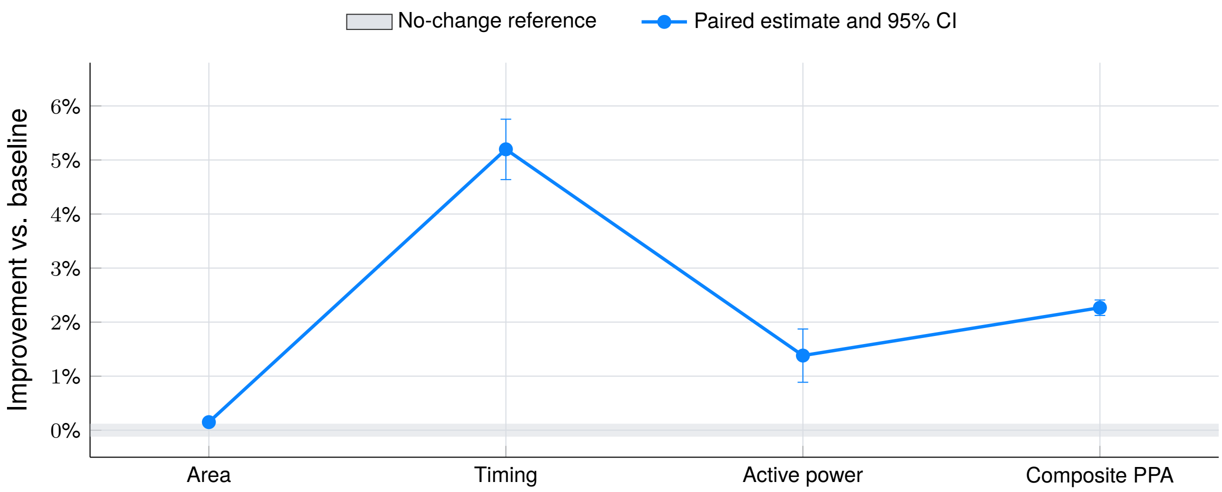


Figure 1. All three primary metric intervals remain on the improvement side of zero.

The composite improves in all 64 paired comparisons. Timing improves in all 64; area records 30 wins, 28 ties and 6 losses; active power records 46 wins and 18 losses. The statistical result belongs to the combined RTL revision; no exact per-transformation PPA attribution is claimed.

The identical area medians do not contradict the positive paired estimate: MWTA is quantized, while the same-seed ratios preserve 30 small wins, 28 ties and 6 losses that an unpaired median comparison would hide.

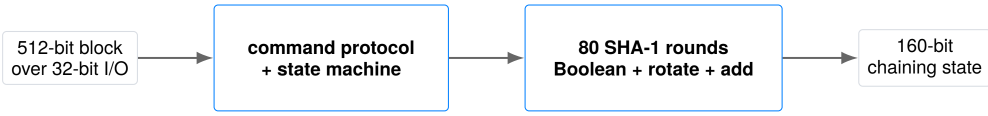
2. Module and benchmark

WHAT IS BEING OPTIMIZED

The evaluated module is a sequential implementation of SHA-1 compression from the public Verilog-to-Routing repository. The exact upstream source is the pinned `sha.v` at commit `95f5c6de…`.

It consumes a 512-bit message block through a 32-bit command/data interface and updates a 160-bit chaining state through 80 rounds of Boolean mixing, rotations, a message schedule and modulo-2^{32} addition. A legal block occupies 80 busy cycles.

```
module sha1(clk_i, rst_i, text_i, text_o, cmd_i, cmd_w_i, cmd_o);
  input clk_i, rst_i, cmd_w_i;
  input  [31:0] text_i;
  output [31:0] text_o;
  input  [2:0]  cmd_i;
  output [3:0]  cmd_o;
endmodule
```



Only `sha.v` may change. Reset behavior, command protocol, digest read order, latency, throughput and every observable cycle remain fixed. SHA-1 is used here as a legacy compute benchmark, not recommended as a security primitive.

Corrected reference and provenance

The upstream artifact does not produce the standard digest for `abc`. A separate conformance correction created the golden source:

$$\mathrm{SHA1}(\texttt{abc}) = \texttt{a9993e364706816aba3e25717850c26c9cd0d89d}.$$

That historical correction is excluded from the optimization claim. Every ratio in this report compares accepted RTL with this corrected, frozen baseline.

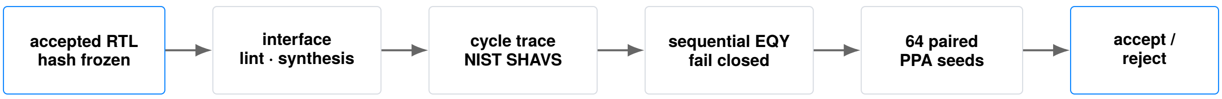
Artifact	SHA-256
Corrected baseline `sha.v`	`191a4f2148a4efda7aadd24480eb13d78a1d2c0c7e8a3fcc37c44f6a8e8011e5`
Accepted `sha.v`	`ea63b97faf69b14409a3077cd3357205c0e044b277d64537b98a4a4773a4f2e7`

3. Evaluation contract

FROZEN BEFORE MEASUREMENT

A PPA comparison is meaningful only when baseline and accepted RTL share the same functional, physical and statistical contract.

Contract dimension	Frozen choice
Editable artifact	`sha.v` only; submission frozen by SHA-256.
Functional behavior	Exact interface, reset, protocol, latency, output order and cycle behavior.
Conformance	NIST SHA-1 Short and Long Message corpus; 129 cases.
Formal	EQY commit `6734d8c2…`; fail closed; 2 CPU, 7 GB, 10 min.
Physical flow	VTR/VPR commit `95f5c6de…`; pinned Linux/amd64 image.
Architecture	`k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml`.
Power	VTR PTM45, 0.9 V, 85 °C; frozen active and idle traces.
PPA sample	64 fixed, paired VPR seeds for certification.
Primary metrics	Total MWTA, critical-path delay and active total power.
Composite	Equal-weight geometric mean of paired area, delay and power ratios.



> Correctness precedes PPA. A failure, timeout or inconclusive formal result invalidates the source before its physical score can qualify it.

The RTL cannot alter testbenches, activity, architecture, tool flags, seeds or parsers. Acceptance belongs to the evaluator, not to the mechanism that proposed the source change.

4. Verification before PPA

EVIDENCE FOR THE ACCEPTED RTL

Gate	What it protects	Status
Structural policy	Interface, editable-file boundary, synthesizability and warning policy	PASS
Cycle regression	Reset, protocol, state progression, latency and output order	PASS
NIST SHAVS	129 Short and Long Message SHA-1 conformance cases	PASS
Sequential EQY	All compared outputs for all cycles after reset under the declared model	PASS
Physical integrity	Route, timing, power, hashes and exact seed pairing	PASS

Formal scope

EQY compares public outputs and stable internal cut points. The proof strategy is conservative and structure-aware: it can reject a legal but structurally distant redesign that does not converge within the laptop resource envelope. That does not weaken a reported pass. The accepted RTL was proved equivalent over the complete sequential circuit under the declared reset model.

Verifier qualification

The 500 deterministic MCY mutations qualify the sensitivity of the verification stack. They are not comparisons among proposed solutions and are not part of the PPA result.

MCY outcome	Count
Detected by simulation	454
Rejected only by formal	28
Proved equivalent	18
Inconclusive	0

All 482 functionally distinguishable mutations were rejected: 454 by simulation and 28 only by formal. The remaining 18 were formally proved equivalent. This shows that formal verification contributes detections beyond the functional corpus; mutation testing does not itself prove absence of every possible defect.

Evidence identity. The accepted formal PASS marker has SHA-256 `9aeb6fa6171b28a9ae33345754b973b407317736d6ac333dd1e4a45d07c575be`.

5. Exact RTL diff

CORRECTED BASELINE TO ACCEPTED SHA.V

The revision changes Boolean representation, shares an internal core and removes a redundant digest concatenation. No register, state transition, port, command, latency or output ordering changes.

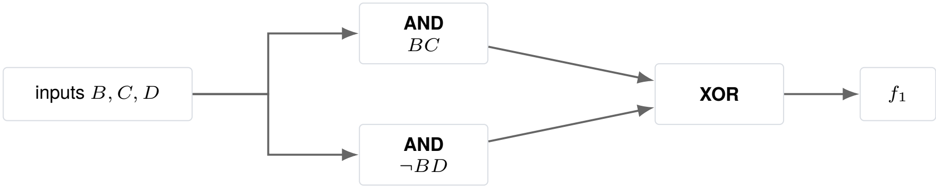
```diff
--- rtl/baseline/sha.v
+++ rtl/accepted/sha.v
@@ -121,20 +121,23 @@
        end

        // Hash functions
-       wire [31:0] SHA1_f1_BCD,SHA1_f2_BCD,SHA1_f3_BCD,SHA1_Wt_1;
-       wire [31:0] SHA1_ft_BCD;
+       wire [31:0] SHA1_B_xor_D,SHA1_C_xor_D;
+       wire [31:0] SHA1_f1_BCD,SHA1_core_f2,SHA1_core_f3,SHA1_core_ft;
+       wire [31:0] SHA1_Wt_1,SHA1_ft_BCD;
        wire [31:0] next_Wt,next_A,next_C;
-       wire [159:0] SHA1_result;

-       assign SHA1_f1_BCD = (B & C) ^ (~B & D);
-       assign SHA1_f2_BCD = B ^ C ^ D;
-       assign SHA1_f3_BCD = (B & C) ^ (C & D) ^ (B & D);
+       assign SHA1_B_xor_D = B ^ D;
+       assign SHA1_C_xor_D = C ^ D;
+       assign SHA1_f1_BCD = (B & C) | (~B & D);
+       assign SHA1_core_f2 = SHA1_B_xor_D ^ SHA1_C_xor_D;
+       assign SHA1_core_f3 = SHA1_B_xor_D & SHA1_C_xor_D;

        // The state machine value is one-based while executing SHA-1 rounds:
        // round=1 computes algorithm round 0 and round=80 computes round 79.
+       assign SHA1_core_ft = (round <= 7'd40) ? SHA1_core_f2 :
+                               (round <= 7'd60) ? SHA1_core_f3 : SHA1_core_f2;
        assign SHA1_ft_BCD = (round <= 7'd20) ? SHA1_f1_BCD :
-                             (round <= 7'd40) ? SHA1_f2_BCD :
-                             (round <= 7'd60) ? SHA1_f3_BCD : SHA1_f2_BCD;
+                               (D ^ SHA1_core_ft);

        // Odin II doesn't support binary operations inside concatenations presently.
        //assign SHA1_Wt_1 = {W13 ^ W8 ^ W2 ^ W0};
@@ -143,8 +146,6 @@
        assign next_Wt = {SHA1_Wt_1[30:0],SHA1_Wt_1[31]};       // NSA fix added
        assign next_A = {A[26:0],A[31:27]} + SHA1_ft_BCD + E + Kt + Wt;
        assign next_C = {B[1:0],B[31:2]};
-
-       assign SHA1_result   = {A,B,C,D,E};

        assign round_plus_1 = round + 1;

@@ -2156,11 +2157,11 @@
                        if (~busy)
                        begin
                                case (read_counter)
-                                       3'b100: text_o <= SHA1_result[5*32-1:4*32];
-                                       3'b011: text_o <= SHA1_result[4*32-1:3*32];
-                                       3'b010: text_o <= SHA1_result[3*32-1:2*32];
-                                       3'b001: text_o <= SHA1_result[2*32-1:1*32];
-                                       3'b000: text_o <= SHA1_result[1*32-1:0*32];
+                                       3'b100: text_o <= A;
+                                       3'b011: text_o <= B;
+                                       3'b010: text_o <= C;
+                                       3'b001: text_o <= D;
+                                       3'b000: text_o <= E;
                                        default:text_o <= 32'b0;
                                endcase
                                if (|read_counter)
```

Attribution boundary. The certified PPA result belongs to this combined revision. The following algebra explains equivalence and the conceptual topology exposed to synthesis; it does not assign an exact fraction of the measured gain to any one transformation.

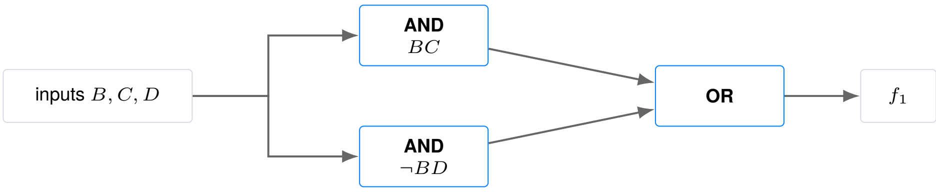
6. Transformation 1: SHA-1 choose function

SOURCE: SHA.V:131 IN ACCEPTED RTL

Corrected baseline · mutually exclusive products joined by XOR



Accepted RTL · the same products joined directly by OR



Why the expressions are equivalent

$$\begin{aligned} F &= BC \oplus (\neg B)D, \\ BC\,(\neg B)D &= 0, \\ \therefore \quad F &= BC \vee (\neg B)D. \end{aligned}$$

The two product terms cannot be one simultaneously: $BC = 1$ requires $B = 1$, whereas $(\neg B)D = 1$ requires $B = 0$. XOR and OR are therefore identical for every input combination. Both forms select C when $B = 1$ and D when $B = 0$.

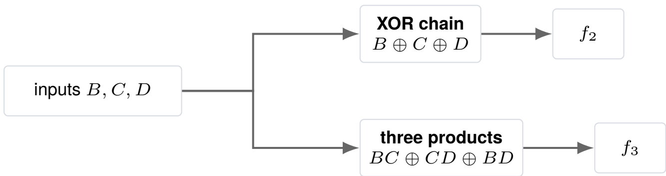
> **Formal result.** EQY proved the actual Verilog implementations equivalent in the complete sequential circuit. The diagram is conceptual; the FPGA mapper chooses the final LUT implementation.

7. Shared SHA-1 Boolean core

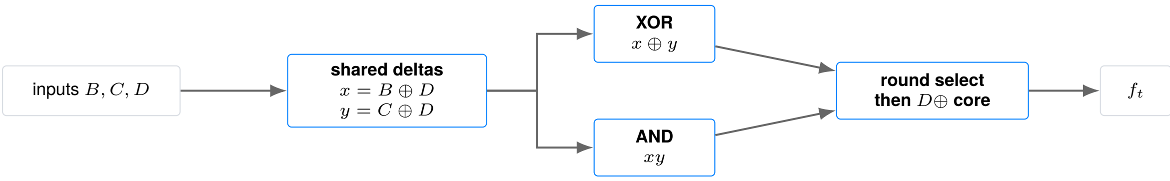
TRANSFORMATION 2 · SOURCE: SHA.V:129–140 IN ACCEPTED RTL

Let $x = B \oplus D$ and $y = C \oplus D$. The accepted RTL computes these two deltas once and reuses them in the parity and majority round families.

Corrected baseline · separate parity and majority networks



Accepted RTL · shared deltas and round-selected core



Why parity is preserved

$$\begin{aligned} D \oplus (x \oplus y) &= D \oplus (B \oplus D) \oplus (C \oplus D) \\ &= B \oplus C \oplus D = f_2. \end{aligned}$$

Why majority is preserved

$$f_3 = D \oplus (xy).$$

If $D = 0$, then $x = B$ and $y = C$, so the expression is BC, exactly the majority of $(B, C, 0)$. If $D = 1$, then $x = \neg B$ and $y = \neg C$, so

$$1 \oplus (\neg B\, \neg C) = B \vee C,$$

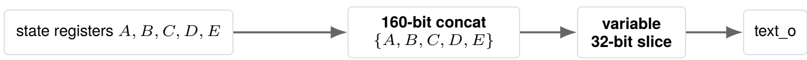
exactly the majority of $(B, C, 1)$. The round selector applies $D \oplus (x \oplus y)$ for parity rounds and $D \oplus (xy)$ for majority rounds, preserving the original SHA-1 schedule.

> **Formal result.** EQY proved the shared implementation and round boundaries equivalent for all reachable cycles after reset.

8. Direct digest readout

TRANSFORMATION 3 · SOURCE: SHA.V:2157–2165 IN ACCEPTED RTL

Corrected baseline · concatenate five state words, then slice the selected word



Accepted RTL · select the existing 32-bit state register directly

Why the observable output is unchanged

The concatenation fixes the exact bit ranges

$$\{A, B, C, D, E\}[159{:}128] = A, \quad \{A, B, C, D, E\}[127{:}96] = B, \quad \ldots \quad , \{A, B, C, D, E\}[31{:}0] = E.$$

Replacing each constant slice by its source register is a direct wiring identity. The same `read_counter` values choose the same word on the same clock edge; no state or latency changes.

> **Formal result.** EQY compared `text_o` on every cycle and proved the direct-register readout equivalent to the concatenation-and-slice form.

9. PPA environment and statistics

OPEN 45 NM FPGA PROXY

The target is VTR's homogeneous `k6_N10_I40_Fi6_L4_frac0_ff1_45nm` architecture: six-input LUTs clustered ten per logic block, with the architecture's routing and timing model. Power uses VTR PTM45 properties at 0.9 V and 85 °C. MWTA is a minimum-width-transistor-area abstraction, not physical square-micrometer area.

Two 5,000-cycle ACE traces are frozen and hashed. The active trace continuously executes diverse legal blocks and completes 60 blocks; the idle trace keeps the clock active after reset with an inactive interface. The same placement and routing are reused for active and idle power analysis.

For every seed s, the primary score is

$$r_s = \sqrt[3]{\frac{A_{c,s}}{A_{b,s}}\,\frac{D_{c,s}}{D_{b,s}}\,\frac{P_{c,s}}{P_{b,s}}}, \qquad \widehat{r} = \exp\left(\frac{1}{64}\sum_{s=1}^{64}\log r_s\right).$$

Here A is routed total area, D critical-path delay and P active total power. Subscripts b and c identify baseline and accepted RTL. The public score is $\widehat{r} = 0.977335$; lower is better and the baseline is 1.0 by construction.

Two-sided 95% Student-t intervals use 63 degrees of freedom in paired log-ratio space. Pairing removes seed-level common variation and answers the decision question directly: what changes when only the RTL changes under the same implementation seed?

Decision rule. Every correctness and integrity gate must pass; at least one primary metric must improve materially; no primary metric may show statistical evidence of regression; and the composite one-sided 95% upper bound must remain below 1.0. The accepted RTL passes all gates.

Energy is informative, not part of the score

Energy per block is derived from active power and routed execution time:

$$E_{\text{block}} = \frac{P_{\text{active,total}}\, t_{\text{workload}}}{60 \text{ completed blocks}}.$$

It remains valuable for interpretation, but the primary score deliberately uses power so that the reported PPA is area–performance–power rather than area–performance–energy.

10. Every paired certification seed

DISTRIBUTION, NOT A SINGLE IMPLEMENTATION

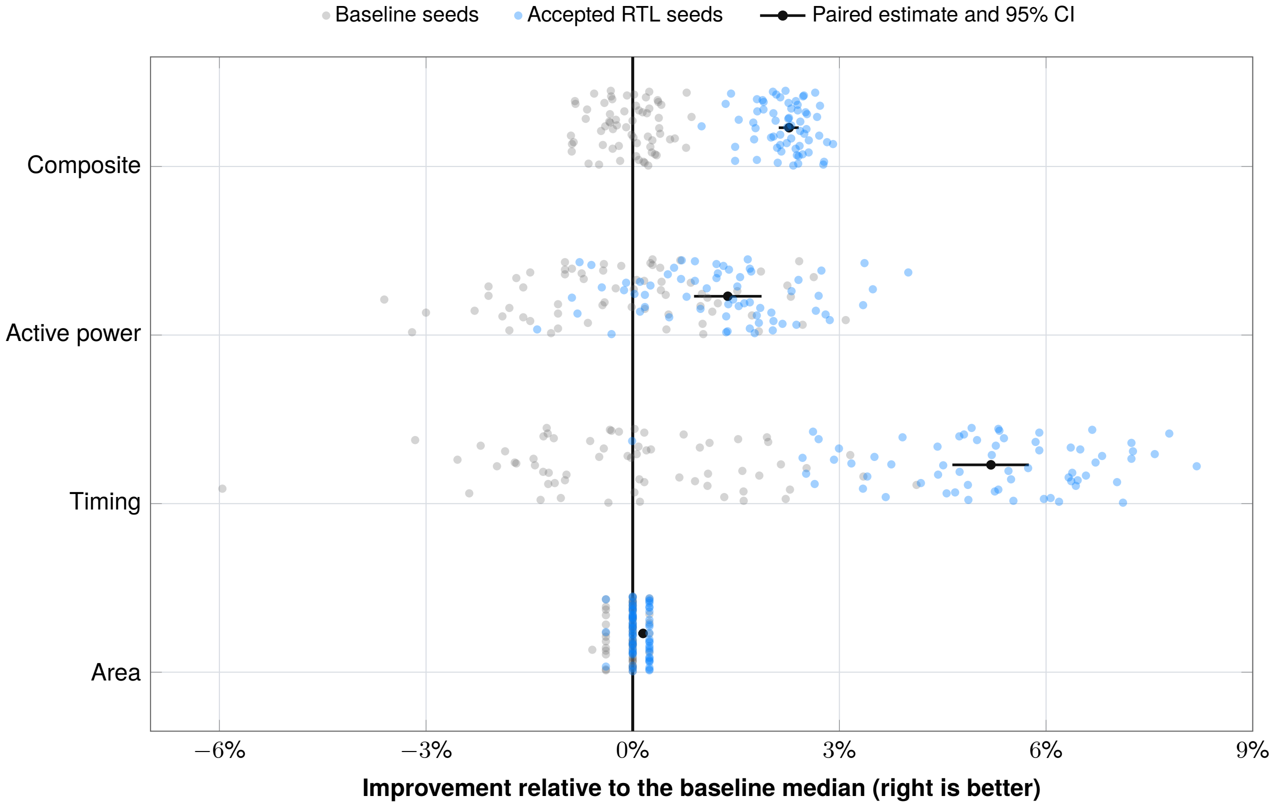


FIGURE 2 · OVERLAID BASELINE AND ACCEPTED SEED CLOUDS

Gray and blue points are absolute seed results normalized to the same baseline median; the same seed receives the same vertical jitter. Black points and segments are the same-seed paired estimate and 95% interval. The clouds show implementation variability; the paired statistic determines confidence.

Primary metric	Wins	Ties	Losses
Area	30	28	6
Timing	64	0	0
Active power	46	0	18
Composite	64	0	0

11. Physical outcome and evidence identity

WHAT CHANGED AFTER IMPLEMENTATION

Metric	Baseline median	Accepted median	Interpretation
Packed CLBs	188	182	6 fewer logic blocks
Registers	892	892	unchanged
Critical path	15.0054 ns	14.2090 ns	5.20% paired improvement
Fmax	66.6426 MHz	70.3781 MHz	higher throughput ceiling
Active total power	9.9125 mW	9.7755 mW	1.38% paired improvement
Idle total power	4.4670 mW	4.3490 mW	informative only
Energy / block	12.3896 nJ	11.5718 nJ	6.51% paired improvement

The source change does not remove architectural state: both implementations retain 892 registers and the same 80-cycle behavior. The accepted source exposes a Boolean topology that VTR packs into six fewer CLBs at the median and routes with a shorter critical path. Lower active power and lower execution time combine to reduce the secondary energy-per-block estimate.

The mechanism statement is deliberately bounded. The Boolean and wiring identities explain why the source is safe; EQY proves the complete sequential equivalence; the 64 routed implementations establish the measured PPA effect. Assigning a precise percentage of gain to an individual source expression would require controlled ablation certifications and is not claimed.

Evidence identity

Artifact	SHA-256
Corrected baseline `sha.v`	`191a4f2148a4efda7aadd24480eb13d78a1d2c0c7e8a3fcc37c44f6a8e8011e5`
Accepted `sha.v`	`ea63b97faf69b14409a3077cd3357205c0e044b277d64537b98a4a4773a4f2e7`
Authoritative certification result	`901488d7018239aa5a498b3710d98299f46d9288a96ab1208e33c3a42284fc46`
EQY PASS marker	`9aeb6fa6171b28a9ae33345754b973b407317736d6ac333dd1e4a45d07c575be`

The compact repository certificate retains all 64 paired metric rows and recomputable statistics. Complete functional, formal, synthesis, route and power logs are retained outside normal Git history and are available for technical audit.

12. Reproducibility and limits

WHAT THE RESULT ESTABLISHES

The fast public audit needs Python 3 only:

```
python3 verify.py
```

It verifies published hashes, reconstructs the exact RTL patch and recomputes medians, paired log-ratio estimates, confidence intervals, win/tie/loss counts and the area-delay-power score from all 64 seed pairs.

The qualified measurement environment is a pinned Linux/amd64 container capped at two CPUs and 7 GB RAM, with runtime networking disabled. Host contention can change wall time but not source hashes, fixed seeds or parsed implementation results.

This report does not provide

- ASIC synthesis, place-and-route or signoff;
- commercial-FPGA implementation data;
- extracted parasitics, PVT corners, clock-tree signoff, IR drop, EM, DRC/LVS, package, yield or production-test analysis;
- proof that the same source edits improve another architecture, technology, workload or tool version;
- exact PPA attribution to an individual transformation;
- a recommendation to use SHA-1 in a new security design.

The 64-seed confidence interval models implementation-seed variability. It does not model process, voltage, temperature, workload or tool-version uncertainty.

Transfer to a customer pilot. Replace the academic proxy with customer-owned RTL, formal strategy, libraries, constraints, activity, tools and signoff policy. The transferable capability is the combination of an inspectable source change and an evaluator that can justify acceptance to the engineers responsible for the system.

Primary references

- VTR/VPR documentation
- VTR power estimation
- EQY sequential equivalence checking
- NIST secure hashing validation
- Reproducible case-study repository

Conclusion. Within the declared VTR 45 nm FPGA proxy, the accepted RTL is cycle-equivalent to the corrected baseline and improves the certified primary PPA estimate by 2.27%. The result is inspectable, statistically supported and explicitly bounded to the evaluation environment.